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Surgical instrument having force feedback capabilities

Abstract

A surgical instrument. The surgical instrument has a force delivery system which communicates a changing force to a user of the surgical instrument when a limit of a surgical function is reached. A sensor senses position of a moving element to communicate a signal to the force delivery system. In various embodiments, the system generates a vibratory force to signal the user.

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Primary Examiner: Jallow; Eyamindae C

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 18/117,130, entitled SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES, filed Mar. 3, 2023, which issued on Apr. 2, 2024 as U.S. Pat. No. 11,944,299, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/334,878, entitled SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES, filed May 31, 2021, which issued on May 16, 2023 as U.S. Pat. No. 11,648,008, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/160,460, entitled SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES, filed Oct. 15, 2018, which issued on Jun. 1, 2021 as U.S. Pat. No. 11,020,113, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 15/223,791, entitled SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES, filed Jul. 29, 2016, which issued on Oct. 16, 2018 as U.S. Pat. No. 10,098,636, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 13/712,090, entitled SURGICAL INSTRUMENT HAVING FORCE FEEDBACK CAPABILITIES, filed Dec. 12, 2012, which issued on Sep. 13, 2016 as U.S. Pat. No. 9,439,649, the entire disclosures of which are hereby incorporated by reference herein. The entire disclosures of U.S. patent application Ser. No. 13/629,702, entitled SURGICAL INSTRUMENT HAVING RECORDING CAPABILITIES, filed Sep. 28, 2012, now U.S. Patent Application Publication No. 2013/0087597, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 13/424,648, entitled SURGICAL INSTRUMENT HAVING RECORDING CAPABILITIES, filed Mar. 20, 2012, which issued on Jun. 17, 2014 as U.S. Pat. No. 8,752,747, which is a divisional application claiming priority under 35 U.S.C. § 121 to U.S. patent application Ser. No. 12/949,099, entitled SURGICAL INSTRUMENT HAVING RECORDING CAPABILITIES, filed Nov. 18, 2010, which issued on May 1, 2012 as U.S. Pat. No. 8,167,185, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 11/343,803, entitled SURGICAL INSTRUMENT HAVING RECORDING CAPABILITIES, filed Jan. 31, 2006, which issued on Dec. 7, 2010 as U.S. Pat. No. 7,845,537, are hereby incorporated by reference herein. (2) The present application is related to the following U.S. patent applications filed on Jan. 31, 2006, which are incorporated herein by reference: U.S. patent application Ser. No. 11/343,498, now U.S. Pat. No. 7,766,210, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH USER FEEDBACK SYSTEM; U.S. patent application Ser. No. 11/343,573, now U.S. Pat. No. 7,416,101, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH LOADING FORCE FEEDBACK; U.S. patent application Ser. No. 11/344,035, now U.S. Pat. No. 7,422,139, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH TACTILE POSITION FEEDBACK; U.S. patent application Ser. No. 11/343,447, now U.S. Pat. No. 7,770,775, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH ADAPTIVE USER FEEDBACK; U.S. patent application Ser. No. 11/343,562, now U.S. Pat. No. 7,568,603, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH ARTICULATABLE END EFFECTOR; U.S. patent application Ser. No.

11/344,024, now U.S. Pat. No. 8,186,555, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH MECHANICAL CLOSURE SYSTEM; U.S. patent application Ser. No. 11/343,321, now U.S. Patent Application Publication No. 2007/0175955, entitled SURGICAL CUTTING AND FASTENING INSTRUMENT WITH CLOSURE TRIGGER LOCKING MECHANISM; U.S. patent application Ser. No. 11/343,563, now U.S. Patent Application Publication No. 2007/0175951, entitled GEARING SELECTOR FOR A POWERED SURGICAL CUTTING AND FASTENING STAPLING INSTRUMENT; U.S. patent application Ser. No. 11/344,020, now U.S. Pat. No. 7,464,846, entitled SURGICAL INSTRUMENT HAVING A REMOVABLE BATTERY; U.S. patent application Ser. No. 11/343,439, now U.S. Pat. No. 7,644,848, entitled ELECTRONIC LOCKOUTS AND SURGICAL INSTRUMENT INCLUDING SAME; U.S. patent application Ser. No. 11/343,547, now U.S. Pat. No. 7,753,904, entitled ENDOSCOPIC SURGICAL INSTRUMENT WITH A HANDLE THAT CAN ARTICULATE WITH RESPECT TO THE SHAFT; U.S. patent application Ser. No. 11/344,021, now U.S. Pat. No. 7,464,849, entitled ELECTRO-MECHANICAL SURGICAL CUTTING AND FASTENING INSTRUMENT HAVING A ROTARY FIRING AND CLOSURE SYSTEM WITH PARALLEL CLOSURE AND ANVIL ALIGNMENT COMPONENTS; U.S. patent application Ser. No. 11/343,546, now U.S. Patent Application Publication No. 2007/0175950, entitled DISPOSABLE STAPLE CARTRIDGE HAVING AN ANVIL WITH TISSUE LOCATOR FOR USE WITH A SURGICAL CUTTING AND FASTENING INSTRUMENT AND MODULAR END EFFECTOR SYSTEM THEREFOR; and U.S. patent application Ser. No. 11/343,545, now U.S. Pat. No. 8,708,213, entitled SURGICAL INSTRUMENT HAVING A FEEDBACK SYSTEM.

BACKGROUND

- (1) The present invention relates in general to surgical instruments, and more particularly to minimally invasive surgical instruments capable of delivering a force feedback to the user.
- (2) Endoscopic surgical instruments are often preferred over traditional open surgical devices because a smaller incision tends to reduce the post-operative recovery time and complications. Consequently, significant development has gone into a range of endoscopic surgical instruments that are suitable for precise placement of a distal end effector at a desired surgical site through a cannula of a trocar. These distal end effectors engage the tissue in a number of ways to achieve a diagnostic or therapeutic effect (e.g., endocutter, grasper, cutter, staplers, clip applier, access device, drug/gene therapy delivery device, and energy device using ultrasound, RF, laser, etc.).
- (3) Known surgical staplers include an end effector that simultaneously makes a longitudinal incision in tissue and applies lines of staples on opposing sides of the incision. The end effector includes a pair of cooperating jaw members that, if the instrument is intended for endoscopic or laparoscopic applications, are capable of passing through a cannula passageway. One of the jaw members receives a staple cartridge having at least two laterally spaced rows of staples. The other jaw member defines an anvil having staple-forming pockets aligned with the rows of staples in the cartridge. The instrument includes a plurality of reciprocating wedges which, when driven distally, pass through openings in the staple cartridge and engage drivers supporting the staples to effect the firing of the staples toward the anvil.
- (4) An example of a surgical stapler suitable for endoscopic applications is described in U.S. Pat. No. 5,465,895, entitled SURGICAL STAPLER INSTRUMENT, which discloses an endocutter with distinct closing and firing actions. A clinician using this device is able to close the jaw members upon tissue to position the tissue prior to firing. Once the clinician has determined that the jaw members are properly gripping tissue, the clinician can then fire the surgical stapler with a single firing stroke, or multiple firing strokes, depending on the device. Firing the surgical stapler causes severing and stapling of the tissue. The simultaneous severing and stapling avoids complications that may arise when performing such actions sequentially with different surgical tools that respectively only sever and staple.

(5) One specific advantage of being able to close upon tissue before firing is that the clinician is able to verify via an endoscope that the desired location for the cut has been achieved, including a sufficient amount of tissue has been captured between opposing jaws. Otherwise, opposing jaws may be drawn too close together, especially pinching at their distal ends, and thus not effectively forming closed staples in the severed tissue. At the other extreme, an excessive amount of clamped tissue may cause binding and an incomplete firing.

(6) When using endoscopic instruments, surgeons often need to be sure if the instrument has completed a function that the surgeon has initiated. For example, the surgeon may initiate articulation of an end-effector of an instrument and allow the articulation to continue to the most extreme articulated position, or an articulation limit. If the surgeon continues to try to articulate the end-effector past the articulation limit, the surgeon will waste valuable time in surgery attempting to complete a function that is past the instrument's limit. Similarly, if a surgeon is to try to fire an already fired instrument, valuable time will be wasted and instrument damage, in some cases, might occur. Therefore, it is advantageous to signal the user of a surgical instrument when the limit of a surgical function is reached.

SUMMARY

(7) In one general aspect, the present invention is directed to a surgical instrument. The surgical instrument has an end effector and a trigger in communication with the end effector. The surgical instrument also has a sensor which communicates with a force delivery system when a limit point of a surgical function has been reached.

(8) Also, in various embodiments, the sensor senses the position of an actuator which drives the surgical function. The actuator may translate or rotate, and may drive one or more of the various functions of the surgical instrument.

(9) In various embodiments, the force deliver system may deliver a vibratory force. The force delivery system may utilize a motor which is driven by a sine wave or a square wave signal to cause the motor to move in a manner that causes vibratory force to signal the user and provide haptic feedback of the limit of a surgical function.

Description

DRAWINGS

(1) Various embodiments of the present invention are described herein by way of example in conjunction with the following figures, wherein

(2) FIGS. 1 and 2 are perspective views of a surgical cutting and fastening instrument according to various embodiments of the present invention;

(3) FIGS. 3-5 are exploded views of an end effector and shaft of the instrument according to various embodiments of the present invention;

(4) FIG. 6 is a side view of the end effector according to various embodiments of the present invention;

(5) FIG. 7 is an exploded view of the handle of the instrument according to various embodiments of the present invention;

(6) FIGS. 8 and 9 are partial perspective views of the handle according to various embodiments of the present invention;

(7) FIG. 10 is a side view of the handle according to various embodiments of the present invention;

(8) FIGS. 10A and 10B illustrate a proportional sensor that may be used according to various embodiments of the present invention;

(9) FIG. 11 is a schematic diagram of a circuit used in the instrument according to various embodiments of the present invention;

(10) FIGS. 12-13 are side views of the handle according to other embodiments of the present

invention;

(11) FIGS. **14-22** illustrate different mechanisms for locking the closure trigger according to various embodiments of the present invention;

(12) FIGS. **23A-B** show a universal joint (“u-joint”) that may be employed at the articulation point of the instrument according to various embodiments of the present invention;

(13) FIGS. **24A-B** shows a torsion cable that may be employed at the articulation point of the instrument according to various embodiments of the present invention;

(14) FIGS. **25-31** illustrate a surgical cutting and fastening instrument with power assist according to another embodiment of the present invention;

(15) FIGS. **32-36** illustrate a surgical cutting and fastening instrument with power assist according to yet another embodiment of the present invention;

(16) FIGS. **37-40** illustrate a surgical cutting and fastening instrument with tactile feedback to embodiments of the present invention;

(17) FIG. **41** illustrates an exploded view of an end effector and shaft of the instrument according to various embodiments of the present invention;

(18) FIG. **42** illustrates a side view of the handle of a mechanically instrument according to various embodiments of the present invention;

(19) FIG. **43** illustrates an exploded view of the handle of the mechanically actuated instrument of FIG. **42**;

(20) FIG. **44** illustrates a block diagram of a recording system for recording various conditions of the instrument according to various embodiments of the present invention;

(21) FIGS. **45-46** illustrate cut away side views of a handle of the instrument showing various sensors according to various embodiments of the present invention;

(22) FIG. **47** illustrates the end effector of the instrument showing various sensors according to various embodiments of the present invention;

(23) FIG. **48** illustrates a firing bar of the instrument including a sensor according to various embodiments of the present invention;

(24) FIG. **49** illustrates a side view of the handle, end effector, and firing bar of the instrument showing a sensor according to various embodiments of the present invention;

(25) FIG. **50** illustrates an exploded view of the staple channel and portions of a staple cartridge of the instrument showing various sensors according to various embodiments of the present invention;

(26) FIG. **51** illustrates a top down view of the staple channel of the instrument showing various sensors according to various embodiments of the present invention;

(27) FIGS. **52A** and **52B** illustrate a flow chart showing a method for operating the instrument according to various embodiments; and

(28) FIG. **53** illustrates a memory chart showing exemplary recorded conditions of the instrument according to various embodiments of the present invention.

(29) FIG. **54** illustrates a graph of voltage applied to a motor versus time showing a changing voltage versus time.

(30) FIG. **55** illustrates a graph of motion of a member versus time

(31) FIG. **56** illustrates a flow chart showing a method of tracking an extent of activation and providing haptic feedback determined by position

DETAILED DESCRIPTION

(32) FIGS. **1** and **2** depict a surgical cutting and fastening instrument **10** according to various embodiments of the present invention. The illustrated embodiment is an endoscopic surgical instrument **10** and in general, the embodiments of the instrument **10** described herein are endoscopic surgical cutting and fastening instruments. It should be noted, however, that according to other embodiments of the present invention, the instrument **10** may be a non-endoscopic surgical cutting instrument, such as a laparoscopic instrument.

(33) The surgical instrument **10** depicted in FIGS. **1** and **2** comprises a handle **6**, a shaft **8**, and an

articulating end effector **12** pivotally connected to the shaft **8** at an articulation pivot **14**. An articulation control **16** may be provided adjacent to the handle **6** to effect rotation of the end effector **12** about the articulation pivot **14**. It will be appreciated that various embodiments may include a non-pivoting end effector, and therefore may not have an articulation pivot **14** or articulation control **16**. Also, in the illustrated embodiment, the end effector **12** is configured to act as an endocutter for clamping, severing and stapling tissue, although, in other embodiments, different types of end effectors may be used, such as end effectors for other types of surgical devices, such as graspers, cutters, staplers, clip applicators, access devices, drug/gene therapy devices, ultrasound, RF or laser devices, etc.

(34) The handle **6** of the instrument **10** may include a closure trigger **18** and a firing trigger **20** for actuating the end effector **12**. It will be appreciated that instruments having end effectors directed to different surgical tasks may have different numbers or types of triggers or other suitable controls for operating the end effector **12**. The end effector **12** is shown separated from the handle **6** by a preferably elongate shaft **8**. In one embodiment, a clinician or operator of the instrument **10** may articulate the end effector **12** relative to the shaft **8** by utilizing the articulation control **16**, as described in more detail in U.S. patent application Ser. No. 11/329,020, entitled SURGICAL INSTRUMENT HAVING AN ARTICULATING END EFFECTOR, now U.S. Pat. No. 7,670,334, which is incorporated herein by reference.

(35) The end effector **12** includes in this example, among other things, a staple channel **22** and a pivotally translatable clamping member, such as an anvil **24**, which are maintained at a spacing that assures effective stapling and severing of tissue clamped in the end effector **12**. The handle **6** includes a pistol grip **26** toward which a closure trigger **18** is pivotally drawn by the clinician to cause clamping or closing of the anvil **24** towards the staple channel **22** of the end effector **12** to thereby clamp tissue positioned between the anvil **24** and channel **22**. The firing trigger **20** is farther outboard of the closure trigger **18**. Once the closure trigger **18** is locked in the closure position as further described below, the firing trigger **20** may rotate slightly toward the pistol grip **26** so that it can be reached by the operator using one hand. Then the operator may pivotally draw the firing trigger **20** toward the pistol grip **26** to cause the stapling and severing of clamped tissue in the end effector **12**. In other embodiments, different types of clamping members besides the anvil **24** could be used, such as, for example, an opposing jaw, etc.

(36) It will be appreciated that the terms “proximal” and “distal” are used herein with reference to a clinician gripping the handle **6** of an instrument **10**. Thus, the end effector **12** is distal with respect to the more proximal handle **6**. It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical” and “horizontal” are used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and absolute.

(37) The closure trigger **18** may be actuated first. Once the clinician is satisfied with the positioning of the end effector **12**, the clinician may draw back the closure trigger **18** to its fully closed, locked position proximate to the pistol grip **26**. The firing trigger **20** may then be actuated. The firing trigger **20** returns to the open position (shown in FIGS. **1** and **2**) when the clinician removes pressure, as described more fully below. A release button on the handle **6**, when depressed may release the locked closure trigger **18**. The release button may be implemented in various forms such as, for example, release button **30** shown in FIGS. **42-43**, slide release button **160** shown in FIG. **14**, and/or button **172** shown in FIG. **16**.

(38) FIGS. **3-6** show embodiments of a rotary-driven end effector **12** and shaft **8** according to various embodiments. FIG. **3** is an exploded view of the end effector **12** according to various embodiments. As shown in the illustrated embodiment, the end effector **12** may include, in addition to the previously-mentioned channel **22** and anvil **24**, a cutting instrument **32**, a sled **33**, a staple cartridge **34** that is removably seated in the channel **22**, and a helical screw shaft **36**. The cutting instrument **32** may be, for example, a knife. The anvil **24** may be pivotally opened and closed at

pivot pins **25** connected to the proximate end of the channel **22**. The anvil **24** may also include a tab **27** at its proximate end that is inserted into a component of the mechanical closure system (described further below) to open and close the anvil **24**. When the closure trigger **18** is actuated, that is, drawn in by a user of the instrument **10**, the anvil **24** may pivot about the pivot pins **25** into the clamped or closed position. If clamping of the end effector **12** is satisfactory, the operator may actuate the firing trigger **20**, which, as explained in more detail below, causes the knife **32** and sled **33** to travel longitudinally along the channel **22**, thereby cutting tissue clamped within the end effector **12**. The movement of the sled **33** along the channel **22** causes the staples (not shown) of the staple cartridge **34** to be driven through the severed tissue and against the closed anvil **24**, which turns the staples to fasten the severed tissue. In various embodiments, the sled **33** may be an integral component of the cartridge **34**. U.S. Pat. No. 6,978,921, entitled SURGICAL STAPLING INSTRUMENT INCORPORATING AN E-BEAM FIRING MECHANISM, which is incorporated herein by reference, provides more details about such two-stroke cutting and fastening instruments. The sled **33** may be part of the cartridge **34**, such that when the knife **32** retracts following the cutting operation, the sled **33** does not retract.

(39) It should be noted that although the embodiments of the instrument **10** described herein employ an end effector **12** that staples the severed tissue, in other embodiments different techniques for fastening or sealing the severed tissue may be used. For example, end effectors that use RF energy or adhesives to fasten the severed tissue may also be used. U.S. Pat. No. 5,709,680, entitled ELECTROSURGICAL HEMOSTATIC DEVICE, and U.S. Pat. No. 5,688,270, entitled ELECTROSURGICAL HEMOSTATIC DEVICE WITH RECESSED AND/OR OFFSET ELECTRODES, which are incorporated herein by reference, disclose an endoscopic cutting instrument that uses RF energy to seal the severed tissue. U.S. patent application Ser. No. 11/267,811, entitled SURGICAL STAPLING INSTRUMENTS STRUCTURED FOR DELIVERY OF MEDICAL AGENTS, now U.S. Pat. No. 7,673,783, and U.S. patent application Ser. No. 11/267,383, entitled SURGICAL STAPLING INSTRUMENTS STRUCTURED FOR PUMP-ASSISTED DELIVERY OF MEDICAL AGENTS, now U.S. Pat. No. 7,607,557, which are also incorporated herein by reference, disclose cutting instruments that uses adhesives to fasten the severed tissue. Accordingly, although the description herein refers to cutting/stapling operations and the like below, it should be recognized that this is an exemplary embodiment and is not meant to be limiting. Other tissue fastening techniques may also be used.

(40) FIGS. **4** and **5** are exploded views and FIG. **6** is a side view of the end effector **12** and shaft **8** according to various embodiments. As shown in the illustrated embodiment, the shaft **8** may include a proximate closure tube **40** and a distal closure tube **42** pivotably linked by a pivot link **44**. The distal closure tube **42** includes an opening **45** into which the tab **27** on the anvil **24** is inserted in order to open and close the anvil **24**, as further described below. Disposed inside the closure tubes **40**, **42** may be a proximate spine tube **46**. Disposed inside the proximate spine tube **46** may be a main rotational (or proximate) drive shaft **48** that communicates with a secondary (or distal) drive shaft **50** via a bevel gear assembly **52**. The secondary drive shaft **50** is connected to a drive gear **54** that engages a proximate drive gear **56** of the helical screw shaft **36**. The vertical bevel gear **52b** may sit and pivot in an opening **57** in the distal end of the proximate spine tube **46**. A distal spine tube **58** may be used to enclose the secondary drive shaft **50** and the drive gears **54**, **56**. Collectively, the main drive shaft **48**, the secondary drive shaft **50**, and the articulation assembly (e.g., the bevel gear assembly **52a-c**) are sometimes referred to herein as the “main drive shaft assembly.”

(41) A bearing **38**, positioned at a distal end of the staple channel **22**, receives the helical drive screw **36**, allowing the helical drive screw **36** to freely rotate with respect to the channel **22**. The helical screw shaft **36** may interface a threaded opening (not shown) of the knife **32** such that rotation of the shaft **36** causes the knife **32** to translate distally or proximally (depending on the direction of the rotation) through the staple channel **22**. Accordingly, when the main drive shaft **48**

is caused to rotate by actuation of the firing trigger **20** (as explained in more detail below), the bevel gear assembly **52a-c** causes the secondary drive shaft **50** to rotate, which in turn, because of the engagement of the drive gears **54, 56**, causes the helical screw shaft **36** to rotate, which causes the knife driving member **32** to travel longitudinally along the channel **22** to cut any tissue clamped within the end effector **12**. The sled **33** may be made of, for example, plastic, and may have a sloped distal surface. As the sled **33** traverses the channel **22**, the sloped forward surface may push up or drive the staples in the staple cartridge through the clamped tissue and against the anvil **24**. The anvil **24** turns the staples, thereby stapling the severed tissue. When the knife **32** is retracted, the knife **32** and sled **33** may become disengaged, thereby leaving the sled **33** at the distal end of the channel **22**.

(42) As described above, because of the lack of user feedback for the cutting/stapling operation, there is a general lack of acceptance among physicians of motor-driven endocutters where the cutting/stapling operation is actuated by merely pressing a button. In contrast, embodiments of the present invention provide a motor-driven endocutter with user-feedback of the deployment, force and/or position of the cutting instrument **32** in end effector **12**.

(43) FIGS. **7-10** illustrate an exemplary embodiment of a motor-driven endocutter, and in particular the handle thereof, that provides user-feedback regarding the deployment and loading force of the cutting instrument **32** in the end effector **12**. In addition, the embodiment may use power provided by the user in retracting the firing trigger **20** to power the device (a so-called “power assist” mode). The embodiment may be used with the rotary driven end effector **12** and shaft **8** embodiments described above. As shown in the illustrated embodiment, the handle **6** includes exterior lower side pieces **59, 60** and exterior upper side pieces **61, 62** that fit together to form, in general, the exterior of the handle **6**. A battery **64**, such as a Li ion battery, may be provided in the pistol grip portion **26** of the handle **6**. The battery **64** powers a motor **65** disposed in an upper portion of the pistol grip portion **26** of the handle **6**. According to various embodiments, the motor **65** may be a DC brushed driving motor having a maximum rotation of, approximately, 5000 RPM. The motor **65** may drive a 90° bevel gear assembly **66** comprising a first bevel gear **68** and a second bevel gear **70**. The bevel gear assembly **66** may drive a planetary gear assembly **72**. The planetary gear assembly **72** may include a pinion gear **74** connected to a drive shaft **76**. The pinion gear **74** may drive a mating ring gear **78** that drives a helical gear drum **80** via a drive shaft **82**. A ring **84** may be threaded on the helical gear drum **80**. Thus, when the motor **65** rotates, the ring **84** is caused to travel along the helical gear drum **80** by means of the interposed bevel gear assembly **66**, planetary gear assembly **72** and ring gear **78**.

(44) The handle **6** may also include a run motor sensor **110** (see FIG. **10**) in communication with the firing trigger **20** to detect when the firing trigger **20** has been drawn in (or “closed”) toward the pistol grip portion **26** of the handle **6** by the operator to thereby actuate the cutting/stapling operation by the end effector **12**. The sensor **110** may be a proportional sensor such as, for example, a rheostat or variable resistor. When the firing trigger **20** is drawn in, the sensor **110** detects the movement, and sends an electrical signal indicative of the voltage (or power) to be supplied to the motor **65**. When the sensor **110** is a variable resistor or the like, the rotation of the motor **65** may be generally proportional to the amount of movement of the firing trigger **20**. That is, if the operator only draws or closes the firing trigger **20** in a little bit, the rotation of the motor **65** is relatively low. When the firing trigger **20** is fully drawn in (or in the fully closed position), the rotation of the motor **65** is at its maximum. In other words, the harder the user pulls on the firing trigger **20**, the more voltage is applied to the motor **65**, causing greater rates of rotation.

(45) The handle **6** may include a middle handle piece **104** adjacent to the upper portion of the firing trigger **20**. The handle **6** also may comprise a bias spring **112** connected between posts on the middle handle piece **104** and the firing trigger **20**. The bias spring **112** may bias the firing trigger **20** to its fully open position. In that way, when the operator releases the firing trigger **20**, the bias spring **112** will pull the firing trigger **20** to its open position, thereby removing actuation of the

sensor **110**, thereby stopping rotation of the motor **65**. Moreover, by virtue of the bias spring **112**, any time a user closes the firing trigger **20**, the user will experience resistance to the closing operation, thereby providing the user with feedback as to the amount of rotation exerted by the motor **65**. Further, the operator could stop retracting the firing trigger **20** to thereby remove force from the sensor **100**, to thereby stop the motor **65**. As such, the user may stop the deployment of the end effector **12**, thereby providing a measure of control of the cutting/fastening operation to the operator.

(46) The distal end of the helical gear drum **80** includes a distal drive shaft **120** that drives a ring gear **122**, which mates with a pinion gear **124**. The pinion gear **124** is connected to the main drive shaft **48** of the main drive shaft assembly. In that way, rotation of the motor **65** causes the main drive shaft assembly to rotate, which causes actuation of the end effector **12**, as described above.

(47) The ring **84** threaded on the helical gear drum **80** may include a post **86** that is disposed within a slot **88** of a slotted arm **90**. The slotted arm **90** has an opening **92** its opposite end **94** that receives a pivot pin **96** that is connected between the handle exterior side pieces **59**, **60**. The pivot pin **96** is also disposed through an opening **100** in the firing trigger **20** and an opening **102** in the middle handle piece **104**.

(48) In addition, the handle **6** may include a reverse motor sensor (or end-of-stroke sensor) **130** and a stop motor (or beginning-of-stroke) sensor **142**. In various embodiments, the reverse motor sensor **130** may be a limit switch located at the distal end of the helical gear drum **80** such that the ring **84** threaded on the helical gear drum **80** contacts and trips the reverse motor sensor **130** when the ring **84** reaches the distal end of the helical gear drum **80**. The reverse motor sensor **130**, when activated, sends a signal to the motor **65** to reverse its rotation direction, thereby withdrawing the knife **32** of the end effector **12** following the cutting operation.

(49) The stop motor sensor **142** may be, for example, a normally-closed limit switch. In various embodiments, it may be located at the proximate end of the helical gear drum **80** so that the ring **84** trips the switch **142** when the ring **84** reaches the proximate end of the helical gear drum **80**.

(50) In operation, when an operator of the instrument **10** pulls back the firing trigger **20**, the sensor **110** detects the deployment of the firing trigger **20** and sends a signal to the motor **65** to cause forward rotation of the motor **65**, for example, at a rate proportional to how hard the operator pulls back the firing trigger **20**. The forward rotation of the motor **65** in turn causes the ring gear **78** at the distal end of the planetary gear assembly **72** to rotate, thereby causing the helical gear drum **80** to rotate, causing the ring **84** threaded on the helical gear drum **80** to travel distally along the helical gear drum **80**. The rotation of the helical gear drum **80** also drives the main drive shaft assembly as described above, which in turn causes deployment of the knife **32** in the end effector **12**. That is, the knife **32** and sled **33** are caused to traverse the channel **22** longitudinally, thereby cutting tissue clamped in the end effector **12**. Also, the stapling operation of the end effector **12** is caused to happen in embodiments where a stapling-type end effector **12** is used.

(51) By the time the cutting/stapling operation of the end effector **12** is complete, the ring **84** on the helical gear drum **80** will have reached the distal end of the helical gear drum **80**, thereby causing the reverse motor sensor **130** to be tripped, which sends a signal to the motor **65** to cause the motor **65** to reverse its rotation. This in turn causes the knife **32** to retract, and also causes the ring **84** on the helical gear drum **80** to move back to the proximate end of the helical gear drum **80**.

(52) The middle handle piece **104** includes a backside shoulder **106** that engages the slotted arm **90** as best shown in FIGS. **8** and **9**. The middle handle piece **104** also has a forward motion stop **107** that engages the firing trigger **20**. The movement of the slotted arm **90** is controlled, as explained above, by rotation of the motor **65**. When the slotted arm **90** rotates counter clockwise as the ring **84** travels from the proximate end of the helical gear drum **80** to the distal end, the middle handle piece **104** will be free to rotate counter clockwise. Thus, as the user draws in the firing trigger **20**, the firing trigger **20** will engage the forward motion stop **107** of the middle handle piece **104**, causing the middle handle piece **104** to rotate counter clockwise. Due to the backside shoulder **106**

engaging the slotted arm **90**, however, the middle handle piece **104** will only be able to rotate counter clockwise as far as the slotted arm **90** permits. In that way, if the motor **65** should stop rotating for some reason, the slotted arm **90** will stop rotating, and the user will not be able to further draw in the firing trigger **20** because the middle handle piece **104** will not be free to rotate counter clockwise due to the slotted arm **90**.

(53) FIGS. **10A** and **10B** illustrate two states of a variable sensor that may be used as the run motor sensor **110** according to various embodiments of the present invention. The sensor **110** may include a face portion **280**, a first electrode (A) **282**, a second electrode (B) **284**, and a compressible dielectric material **286** between the electrodes **282**, **284**, such as, for example, an electroactive polymer (EAP). The sensor **110** may be positioned such that the face portion **280** contacts the firing trigger **20** when retracted. Accordingly, when the firing trigger **20** is retracted, the dielectric material **286** is compressed, as shown in FIG. **10B**, such that the electrodes **282**, **284** are closer together. Since the distance “b” between the electrodes **282**, **284** is directly related to the impedance between the electrodes **282**, **284**, the greater the distance the more impedance, and the closer the distance the less impedance. In that way, the amount that the dielectric **286** is compressed due to retraction of the firing trigger **20** (denoted as force “F” in FIG. **42**) is proportional to the impedance between the electrodes **282**, **284**, which can be used to proportionally control the motor **65**.

(54) Components of an exemplary closure system for closing (or clamping) the anvil **24** of the end effector **12** by retracting the closure trigger **18** are also shown in FIGS. **7-10**. In the illustrated embodiment, the closure system includes a yoke **250** connected to the closure trigger **18** by a pivot pin **251** inserted through aligned openings in both the closure trigger **18** and the yoke **250**. A pivot pin **252**, about which the closure trigger **18** pivots, is inserted through another opening in the closure trigger **18** which is offset from where the pin **251** is inserted through the closure trigger **18**. Thus, retraction of the closure trigger **18** causes the upper part of the closure trigger **18**, to which the yoke **250** is attached via the pin **251**, to rotate counterclockwise. The distal end of the yoke **250** is connected, via a pin **254**, to a first closure bracket **256**. The first closure bracket **256** connects to a second closure bracket **258**. Collectively, the closure brackets **256**, **258** define an opening in which the proximate end of the proximate closure tube **40** (see FIG. **4**) is seated and held such that longitudinal movement of the closure brackets **256**, **258** causes longitudinal motion by the proximate closure tube **40**. The instrument **10** also includes a closure rod **260** disposed inside the proximate closure tube **40**. The closure rod **260** may include a window **261** into which a post **263** on one of the handle exterior pieces, such as exterior lower side piece **59** in the illustrated embodiment, is disposed to fixedly connect the closure rod **260** to the handle **6**. In that way, the proximate closure tube **40** is capable of moving longitudinally relative to the closure rod **260**. The closure rod **260** may also include a distal collar **267** that fits into a cavity **269** in proximate spine tube **46** and is retained therein by a cap **271** (see FIG. **4**).

(55) In operation, when the yoke **250** rotates due to retraction of the closure trigger **18**, the closure brackets **256**, **258** cause the proximate closure tube **40** to move distally (i.e., away from the handle end of the instrument **10**), which causes the distal closure tube **42** to move distally, which causes the anvil **24** to rotate about the pivot pins **25** into the clamped or closed position. When the closure trigger **18** is unlocked from the locked position, the proximate closure tube **40** is caused to slide proximally, which causes the distal closure tube **42** to slide proximally, which, by virtue of the tab **27** being inserted in the window **45** of the distal closure tube **42**, causes the anvil **24** to pivot about the pivot pins **25** into the open or unclamped position. In that way, by retracting and locking the closure trigger **18**, an operator may clamp tissue between the anvil **24** and channel **22**, and may unclamp the tissue following the cutting/stapling operation by unlocking the closure trigger **20** from the locked position.

(56) FIG. **11** is a schematic diagram of an electrical circuit of the instrument **10** according to various embodiments of the present invention. When an operator initially pulls in the firing trigger

20 after locking the closure trigger 18, the sensor 110 is activated, allowing current to flow there through. If the normally-open reverse motor sensor switch 130 is open (meaning the end of the end effector stroke has not been reached), current will flow to a single pole, double throw relay 132. Since the reverse motor sensor switch 130 is not closed, the inductor 134 of the relay 132 will not be energized, so the relay 132 will be in its non-energized state. The circuit also includes a cartridge lockout sensor 136. If the end effector 12 includes a staple cartridge 34, the sensor 136 will be in the closed state, allowing current to flow. Otherwise, if the end effector 12 does not include a staple cartridge 34, the sensor 136 will be open, thereby preventing the battery 64 from powering the motor 65.

(57) When the staple cartridge 34 is present, the sensor 136 is closed, which energizes a single pole, single throw relay 138. When the relay 138 is energized, current flows through the relay 136, through the variable resistor sensor 110, and to the motor 65 via a double pole, double throw relay 140, thereby powering the motor 65 and allowing it to rotate in the forward direction.

(58) When the end effector 12 reaches the end of its stroke, the reverse motor sensor 130 will be activated, thereby closing the switch 130 and energizing the relay 134. This causes the relay 134 to assume its energized state (not shown in FIG. 13), which causes current to bypass the cartridge lockout sensor 136 and variable resistor 110, and instead causes current to flow to both the normally-closed double pole, double throw relay 142 and back to the motor 65, but in a manner, via the relay 140, that causes the motor 65 to reverse its rotational direction.

(59) Because the stop motor sensor switch 142 is normally-closed, current will flow back to the relay 134 to keep it closed until the switch 142 opens. When the knife 32 is fully retracted, the stop motor sensor switch 142 is activated, causing the switch 142 to open, thereby removing power from the motor 65.

(60) In other embodiments, rather than a proportional-type sensor 110, an on-off type sensor could be used. In such embodiments, the rate of rotation of the motor 65 would not be proportional to the force applied by the operator. Rather, the motor 65 would generally rotate at a constant rate. But the operator would still experience force feedback because the firing trigger 20 is geared into the gear drive train.

(61) FIG. 12 is a side-view of the handle 6 of a power-assist motorized endocutter according to another embodiment. The embodiment of FIG. 12 is similar to that of FIGS. 7-10 except that in the embodiment of FIG. 12, there is no slotted arm connected to the ring 84 threaded on the helical gear drum 80. Instead, in the embodiment of FIG. 12, the ring 84 includes a sensor portion 114 that moves with the ring 84 as the ring 84 advances down (and back) on the helical gear drum 80. The sensor portion 114 includes a notch 116. The reverse motor sensor 130 may be located at the distal end of the notch 116 and the stop motor sensor 142 may be located at the proximate end of the notch 116. As the ring 84 moves down the helical gear drum 80 (and back), the sensor portion 114 moves with it. Further, as shown in FIG. 12, the middle piece 104 may have an arm 118 that extends into the notch 12.

(62) In operation, as an operator of the instrument 10 retracts in the firing trigger 20 toward the pistol grip 26, the run motor sensor 110 detects the motion and sends a signal to power the motor 65, which causes, among other things, the helical gear drum 80 to rotate. As the helical gear drum 80 rotates, the ring 84 threaded on the helical gear drum 80 advances (or retracts, depending on the rotation). Also, due to the pulling in of the firing trigger 20, the middle piece 104 is caused to rotate counter clockwise with the firing trigger 20 due to the forward motion stop 107 that engages the firing trigger 20. The counter clockwise rotation of the middle piece 104 cause the arm 118 to rotate counter clockwise with the sensor portion 114 of the ring 84 such that the arm 118 stays disposed in the notch 116. When the ring 84 reaches the distal end of the helical gear drum 80, the arm 118 will contact and thereby trip the reverse motor sensor 130. Similarly, when the ring 84 reaches the proximate end of the helical gear drum 80, the arm will contact and thereby trip the stop motor sensor 142. Such actions may reverse and stop the motor 65, respectively as described

above.

(63) FIG. 13 is a side-view of the handle 6 of a power-assist motorized endocutter according to another embodiment. The embodiment of FIG. 13 is similar to that of FIGS. 7-10 except that in the embodiment of FIG. 13, there is no slot in the arm 90. Instead, the ring 84 threaded on the helical gear drum 80 includes a vertical channel 126. Instead of a slot, the arm 90 includes a post 128 that is disposed in the channel 126. As the helical gear drum 80 rotates, the ring 84 threaded on the helical gear drum 80 advances (or retracts, depending on the rotation). The arm 90 rotates counter clockwise as the ring 84 advances due to the post 128 being disposed in the channel 126, as shown in FIG. 13.

(64) As mentioned above, in using a two-stroke motorized instrument, the operator first pulls back and locks the closure trigger 18. FIGS. 14 and 15 show one embodiment of a way to lock the closure trigger 18 to the pistol grip portion 26 of the handle 6. In the illustrated embodiment, the pistol grip portion 26 includes a hook 150 that is biased to rotate counter clockwise about a pivot point 151 by a torsion spring 152. Also, the closure trigger 18 includes a closure bar 154. As the operator draws in the closure trigger 18, the closure bar 154 engages a sloped portion 156 of the hook 150, thereby rotating the hook 150 upward (or clockwise in FIGS. 14-15) until the closure bar 154 completely passes the sloped portion 156 passes into a recessed notch 158 of the hook 150, which locks the closure trigger 18 in place. The operator may release the closure trigger 18 by pushing down on a slide button release 160 on the back or opposite side of the pistol grip portion 26. Pushing down the slide button release 160 rotates the hook 150 clockwise such that the closure bar 154 is released from the recessed notch 158.

(65) FIG. 16 shows another closure trigger locking mechanism according to various embodiments. In the embodiment of FIG. 16, the closure trigger 18 includes a wedge 160 having an arrow-head portion 161. The arrow-head portion 161 is biased downward (or clockwise) by a leaf spring 162. The wedge 160 and leaf spring 162 may be made from, for example, molded plastic. When the closure trigger 18 is retracted, the arrow-head portion 161 is inserted through an opening 164 in the pistol grip portion 26 of the handle 6. A lower chamfered surface 166 of the arrow-head portion 161 engages a lower sidewall 168 of the opening 164, forcing the arrow-head portion 161 to rotate counter clockwise. Eventually the lower chamfered surface 166 fully passes the lower sidewall 168, removing the counter clockwise force on the arrow-head portion 161, causing the lower sidewall 168 to slip into a locked position in a notch 170 behind the arrow-head portion 161.

(66) To unlock the closure trigger 18, a user presses down on a button 172 on the opposite side of the closure trigger 18, causing the arrow-head portion 161 to rotate counter clockwise and allowing the arrow-head portion 161 to slide out of the opening 164.

(67) FIGS. 17-22 show a closure trigger locking mechanism according to another embodiment. As shown in this embodiment, the closure trigger 18 includes a flexible longitudinal arm 176 that includes a lateral pin 178 extending therefrom. The arm 176 and pin 178 may be made from molded plastic, for example. The pistol grip portion 26 of the handle 6 includes an opening 180 with a laterally extending wedge 182 disposed therein. When the closure trigger 18 is retracted, the pin 178 engages the wedge 182, and the pin 178 is forced downward (i.e., the arm 176 is rotated clockwise) by the lower surface 184 of the wedge 182, as shown in FIGS. 17 and 18. When the pin 178 fully passes the lower surface 184, the clockwise force on the arm 176 is removed, and the pin 178 is rotated counter clockwise such that the pin 178 comes to rest in a notch 186 behind the wedge 182, as shown in FIG. 19, thereby locking the closure trigger 18. The pin 178 is further held in place in the locked position by a flexible stop 188 extending from the wedge 184.

(68) To unlock the closure trigger 18, the operator may further squeeze the closure trigger 18, causing the pin 178 to engage a sloped backwall 190 of the opening 180, forcing the pin 178 upward past the flexible stop 188, as shown in FIGS. 20 and 21. The pin 178 is then free to travel out an upper channel 192 in the opening 180 such that the closure trigger 18 is no longer locked to the pistol grip portion 26, as shown in FIG. 22.

(69) FIGS. 23A-B show a universal joint (“u-joint”) **195**. The second piece **195-2** of the u-joint **195** rotates in a horizontal plane in which the first piece **195-1** lies. FIG. 23A shows the u-joint **195** in a linear (180°) orientation and FIG. 23B shows the u-joint **195** at approximately a 150° orientation. The u-joint **195** may be used instead of the bevel gears **52a-c** (see FIG. 4, for example) at the articulation point **14** of the main drive shaft assembly to articulate the end effector **12**. FIGS. 24A-B show a torsion cable **197** that may be used in lieu of both the bevel gears **52a-c** and the u-joint **195** to realize articulation of the end effector **12**.

(70) FIGS. 25-31 illustrate another embodiment of a motorized, two-stroke surgical cutting and fastening instrument **10** with power assist according to another embodiment of the present invention. The embodiment of FIGS. 25-31 is similar to that of FIGS. 6-10 except that instead of the helical gear drum **80**, the embodiment of FIGS. 23-28 includes an alternative gear drive assembly. The embodiment of FIGS. 25-31 includes a gear box assembly **200** including a number of gears disposed in a frame **201**, wherein the gears are connected between the planetary gear **72** and the pinion gear **124** at the proximate end of the drive shaft **48**. As explained further below, the gear box assembly **200** provides feedback to the user via the firing trigger **20** regarding the deployment and loading force of the end effector **12**. Also, the user may provide power to the system via the gear box assembly **200** to assist the deployment of the end effector **12**. In that sense, like the embodiments described above, the embodiment of FIGS. 23-32 is another power assist motorized instrument **10** that provides feedback to the user regarding the loading force experienced by the instrument.

(71) In the illustrated embodiment, the firing trigger **20** includes two pieces: a main body portion **202** and a stiffening portion **204**. The main body portion **202** may be made of plastic, for example, and the stiffening portion **204** may be made out of a more rigid material, such as metal. In the illustrated embodiment, the stiffening portion **204** is adjacent to the main body portion **202**, but according to other embodiments, the stiffening portion **204** could be disposed inside the main body portion **202**. A pivot pin **207** may be inserted through openings in the firing trigger pieces **202**, **204** and may be the point about which the firing trigger **20** rotates. In addition, a spring **222** may bias the firing trigger **20** to rotate in a counter clockwise direction. The spring **222** may have a distal end connected to a pin **224** that is connected to the pieces **202**, **204** of the firing trigger **20**. The proximate end of the spring **222** may be connected to one of the handle exterior lower side pieces **59**, **60**.

(72) In the illustrated embodiment, both the main body portion **202** and the stiffening portion **204** includes gear portions **206**, **208** (respectively) at their upper end portions. The gear portions **206**, **208** engage a gear in the gear box assembly **200**, as explained below, to drive the main drive shaft assembly and to provide feedback to the user regarding the deployment of the end effector **12**.

(73) The gear box assembly **200** may include as shown, in the illustrated embodiment, six (6) gears. A first gear **210** of the gear box assembly **200** engages the gear portions **206**, **208** of the firing trigger **20**. In addition, the first gear **210** engages a smaller second gear **212**, the smaller second gear **212** being coaxial with a large third gear **214**. The third gear **214** engages a smaller fourth gear **216**, the smaller fourth gear being coaxial with a fifth gear **218**. The fifth gear **218** is a 90° bevel gear that engages a mating 90° bevel gear **220** (best shown in FIG. 31) that is connected to the pinion gear **124** that drives the main drive shaft **48**.

(74) In operation, when the user retracts the firing trigger **20**, a run motor sensor (not shown) is activated, which may provide a signal to the motor **65** to rotate at a rate proportional to the extent or force with which the operator is retracting the firing trigger **20**. This causes the motor **65** to rotate at a speed proportional to the signal from the sensor. The sensor is not shown for this embodiment, but it could be similar to the run motor sensor **110** described above. The sensor could be located in the handle **6** such that it is depressed when the firing trigger **20** is retracted. Also, instead of a proportional-type sensor, an on/off type sensor may be used.

(75) Rotation of the motor **65** causes the bevel gears **68**, **70** to rotate, which causes the planetary

gear **72** to rotate, which causes, via the drive shaft **76**, the ring gear **122** to rotate. The ring gear **122** meshes with the pinion gear **124**, which is connected to the main drive shaft **48**. Thus, rotation of the pinion gear **124** drives the main drive shaft **48**, which causes actuation of the cutting/stapling operation of the end effector **12**.

(76) Forward rotation of the pinion gear **124** in turn causes the bevel gear **220** to rotate, which causes, by way of the rest of the gears of the gear box assembly **200**, the first gear **210** to rotate. The first gear **210** engages the gear portions **206**, **208** of the firing trigger **20**, thereby causing the firing trigger **20** to rotate counter clockwise when the motor **65** provides forward drive for the end effector **12** (and to rotate counter clockwise when the motor **65** rotates in reverse to retract the end effector **12**). In that way, the user experiences feedback regarding loading force and deployment of the end effector **12** by way of the user's grip on the firing trigger **20**. Thus, when the user retracts the firing trigger **20**, the operator will experience a resistance related to the load force experienced by the end effector **12**. Similarly, when the operator releases the firing trigger **20** after the cutting/stapling operation so that it can return to its original position, the user will experience a clockwise rotation force from the firing trigger **20** that is generally proportional to the reverse speed of the motor **65**.

(77) It should also be noted that in this embodiment the user can apply force (either in lieu of or in addition to the force from the motor **65**) to actuate the main drive shaft assembly (and hence the cutting/stapling operation of the end effector **12**) through retracting the firing trigger **20**. That is, retracting the firing trigger **20** causes the gear portions **206**, **208** to rotate counter clockwise, which causes the gears of the gear box assembly **200** to rotate, thereby causing the pinion gear **124** to rotate, which causes the main drive shaft **48** to rotate.

(78) Although not shown in FIGS. **25-31**, the instrument **10** may further include reverse motor and stop motor sensors. As described above, the reverse motor and stop motor sensors may detect, respectively, the end of the cutting stroke (full deployment of the knife **32**) and the end of retraction operation (full retraction of the knife **32**). A similar circuit to that described above in connection with FIG. **11** may be used to appropriately power the motor **65**.

(79) FIGS. **32-36** illustrate a two-stroke, motorized surgical cutting and fastening instrument **10** with power assist according to another embodiment. The embodiment of FIGS. **32-36** is similar to that of FIGS. **25-31** except that in the embodiment of FIGS. **32-36**, the firing trigger **20** includes a lower portion **228** and an upper portion **230**. Both portions **228**, **230** are connected to and pivot about a pivot pin **207** that is disposed through each portion **228**, **230**. The upper portion **230** includes a gear portion **232** that engages the first gear **210** of the gear box assembly **200**. The spring **222** is connected to the upper portion **230** such that the upper portion is biased to rotate in the clockwise direction. The upper portion **230** may also include a lower arm **234** that contacts an upper surface of the lower portion **228** of the firing trigger **20** such that when the upper portion **230** is caused to rotate clockwise the lower portion **228** also rotates clockwise, and when the lower portion **228** rotates counter clockwise the upper portion **230** also rotates counter clockwise. Similarly, the lower portion **228** includes a rotational stop **238** that engages a shoulder of the upper portion **230**. In that way, when the upper portion **230** is caused to rotate counter clockwise the lower portion **228** also rotates counter clockwise, and when the lower portion **228** rotates clockwise the upper portion **230** also rotates clockwise.

(80) The illustrated embodiment also includes the run motor sensor **110** that communicates a signal to the motor **65** that, in various embodiments, may cause the motor **65** to rotate at a speed proportional to the force applied by the operator when retracting the firing trigger **20**. The sensor **110** may be, for example, a rheostat or some other variable resistance sensor, as explained herein. In addition, the instrument **10** may include reverse motor sensor **130** that is tripped or switched when contacted by a front face **242** of the upper portion **230** of the firing trigger **20**. When activated, the reverse motor sensor **130** sends a signal to the motor **65** to reverse direction. Also, the instrument **10** may include a stop motor sensor **142** that is tripped or actuated when contacted by

the lower portion **228** of the firing trigger **20**. When activated, the stop motor sensor **142** sends a signal to stop the reverse rotation of the motor **65**.

(81) In operation, when an operator retracts the closure trigger **18** into the locked position, the firing trigger **20** is retracted slightly (through mechanisms known in the art, including U.S. Pat. Nos. 6,978,921 and 6,905,057, which are incorporated herein by reference) so that the user can grasp the firing trigger **20** to initiate the cutting/stapling operation, as shown in FIGS. **32** and **33**. At that point, as shown in FIG. **33**, the gear portion **232** of the upper portion **230** of the firing trigger **20** moves into engagement with the first gear **210** of the gear box assembly **200**. When the operator retracts the firing trigger **20**, according to various embodiments, the firing trigger **20** may rotate a small amount, such as five degrees, before tripping the run motor sensor **110**, as shown in FIG. **34**. Activation of the sensor **110** causes the motor **65** to forward rotate at a rate proportional to the retraction force applied by the operator. The forward rotation of the motor **65** causes, as described above, the main drive shaft **48** to rotate, which causes the knife **32** in the end effector **12** to be deployed (i.e., begin traversing the channel **22**). Rotation of the pinion gear **124**, which is connected to the main drive shaft **48**, causes the gears **210-220** in the gear box assembly **200** to rotate. Since the first gear **210** is in engagement with the gear portion **232** of the upper portion **230** of the firing trigger **20**, the upper portion **232** is caused to rotate counter clockwise, which causes the lower portion **228** to also rotate counter clockwise.

(82) When the knife **32** is fully deployed (i.e., at the end of the cutting stroke), the front face **242** of the upper portion **230** trips the reverse motor sensor **130**, which sends a signal to the motor **65** to reverse rotational directional. This causes the main drive shaft assembly to reverse rotational direction to retract the knife **32**. Reverse rotation of the main drive shaft assembly also causes the gears **210-220** in the gear box assembly to reverse direction, which causes the upper portion **230** of the firing trigger **20** to rotate clockwise, which causes the lower portion **228** of the firing trigger **20** to rotate clockwise until the lower portion **228** trips or actuates the stop motor sensor **142** when the knife **32** is fully retracted, which causes the motor **65** to stop. In that way, the user experiences feedback regarding deployment of the end effector **12** by way of the user's grip on the firing trigger **20**. Thus, when the user retracts the firing trigger **20**, the operator will experience a resistance related to the deployment of the end effector **12** and, in particular, to the loading force experienced by the knife **32**. Similarly, when the operator releases the firing trigger **20** after the cutting/stapling operation so that it can return to its original position, the user will experience a clockwise rotation force from the firing trigger **20** that is generally proportional to the reverse speed of the motor **65**.

(83) It should also be noted that in this embodiment the user can apply force (either in lieu of or in addition to the force from the motor **65**) to actuate the main drive shaft assembly (and hence the cutting/stapling operation of the end effector **12**) through retracting the firing trigger **20**. That is, retracting the firing trigger **20** causes the gear portion **232** of the upper portion **230** to rotate counter clockwise, which causes the gears of the gear box assembly **200** to rotate, thereby causing the pinion gear **124** to rotate, which causes the main drive shaft assembly to rotate.

(84) The above-described embodiments employed power-assist user feedback systems, with or without adaptive control (e.g., using a sensor **110**, **130**, and **142** outside of the closed loop system of the motor **65**, gear drive train, and end effector **12**) for a two-stroke, motorized surgical cutting and fastening instrument. That is, force applied by the user in retracting the firing trigger **20** may be added to the force applied by the motor **65** by virtue of the firing trigger **20** being geared into (either directly or indirectly) the gear drive train between the motor **65** and the main drive shaft **48**. In other embodiments of the present invention, the user may be provided with tactile feedback regarding the position of the knife **32** in the end effector, but without having the firing trigger **20** geared into the gear drive train. FIGS. **37-40** illustrate a motorized surgical cutting and fastening instrument with such a tactile position feedback system.

(85) In the illustrated embodiment of FIGS. **37-40**, the firing trigger **20** may have a lower portion **228** and an upper portion **230**, similar to the instrument **10** shown in FIGS. **32-36**. Unlike the

embodiment of FIG. 32-36, however, the upper portion 230 does not have a gear portion that mates with part of the gear drive train. Instead, the instrument includes a second motor 265 with a threaded rod 266 threaded therein. The threaded rod 266 reciprocates longitudinally in and out of the motor 265 as the motor 265 rotates, depending on the direction of rotation. The instrument 10 also includes an encoder 268 that is responsive to the rotations of the main drive shaft 48 for translating the incremental angular motion of the main drive shaft 48 (or other component of the main drive assembly) into a corresponding series of digital signals, for example. In the illustrated embodiment, the pinion gear 124 includes a proximate drive shaft 270 that connects to the encoder 268.

(86) The instrument 10 also includes a control circuit (not shown), which may be implemented using a microcontroller or some other type of integrated circuit, that receives the digital signals from the encoder 268. Based on the signals from the encoder 268, the control circuit may calculate the stage of deployment of the knife 32 in the end effector 12. That is, the control circuit can calculate if the knife 32 is fully deployed, fully retracted, or at an intermittent stage. Based on the calculation of the stage of deployment of the end effector 12, the control circuit may send a signal to the second motor 265 to control its rotation to thereby control the reciprocating movement of the threaded rod 266.

(87) In operation, as shown in FIG. 37, when the closure trigger 18 is not locked into the clamped position, the firing trigger 20 rotated away from the pistol grip portion 26 of the handle 6 such that the front face 242 of the upper portion 230 of the firing trigger 20 is not in contact with the proximate end of the threaded rod 266. When the operator retracts the closure trigger 18 and locks it in the clamped position, the firing trigger 20 rotates slightly towards the closure trigger 20 so that the operator can grasp the firing trigger 20, as shown in FIG. 38. In this position, the front face 242 of the upper portion 230 contacts the proximate end of the threaded rod 266.

(88) As the user then retracts the firing trigger 20, after an initial rotational amount (e.g. 5 degrees of rotation) the run motor sensor 110 may be activated such that, as explained above, the sensor 110 sends a signal to the motor 65 to cause it to rotate at a forward speed proportional to the amount of retraction force applied by the operator to the firing trigger 20. Forward rotation of the motor 65 causes the main drive shaft 48 to rotate via the gear drive train, which causes the knife 32 and sled 33 to travel down the channel 22 and sever tissue clamped in the end effector 12. The control circuit receives the output signals from the encoder 268 regarding the incremental rotations of the main drive shaft assembly and sends a signal to the second motor 265 to cause the second motor 265 to rotate, which causes the threaded rod 266 to retract into the motor 265. This allows the upper portion 230 of the firing trigger 20 to rotate counter clockwise, which allows the lower portion 228 of the firing trigger to also rotate counter clockwise. In that way, because the reciprocating movement of the threaded rod 266 is related to the rotations of the main drive shaft assembly, the operator of the instrument 10, by way of his/her grip on the firing trigger 20, experiences tactile feedback as to the position of the end effector 12. The retraction force applied by the operator, however, does not directly affect the drive of the main drive shaft assembly because the firing trigger 20 is not geared into the gear drive train in this embodiment.

(89) By virtue of tracking the incremental rotations of the main drive shaft assembly via the output signals from the encoder 268, the control circuit can calculate when the knife 32 is fully deployed (i.e., fully extended). At this point, the control circuit may send a signal to the motor 65 to reverse direction to cause retraction of the knife 32. The reverse direction of the motor 65 causes the rotation of the main drive shaft assembly to reverse direction, which is also detected by the encoder 268. Based on the reverse rotation detected by the encoder 268, the control circuit sends a signal to the second motor 265 to cause it to reverse rotational direction such that the threaded rod 266 starts to extend longitudinally from the motor 265. This motion forces the upper portion 230 of the firing trigger 20 to rotate clockwise, which causes the lower portion 228 to rotate clockwise. In that way, the operator may experience a clockwise force from the firing trigger 20, which provides feedback

to the operator as to the retraction position of the knife **32** in the end effector **12**. The control circuit can determine when the knife **32** is fully retracted. At this point, the control circuit may send a signal to the motor **65** to stop rotation.

(90) According to other embodiments, rather than having the control circuit determine the position of the knife **32**, reverse motor and stop motor sensors may be used, as described above. In addition, rather than using a proportional sensor **110** to control the rotation of the motor **65**, an on/off switch or sensor can be used. In such an embodiment, the operator would not be able to control the rate of rotation of the motor **65**. Rather, it would rotate at a preprogrammed rate.

(91) FIGS. **41-43** illustrate an exemplary embodiment of a mechanically actuated endocutter, and in particular the handle **6**, shaft **8** and end effector **12** thereof. Further details of a mechanically actuated endocutter may be found in U.S. patent application Ser. No. 11/052,632 entitled, SURGICAL STAPLING INSTRUMENT INCORPORATING A MULTI-STROKE FIRING MECHANISM WITH AUTOMATIC END OF FIRING TRAVEL RETRACTION, now U.S. Pat. No. 7,083,075, which is incorporated herein by reference. With reference to FIG. **41**, the end effector **12** responds to the closure motion from the handle **6** (not depicted in FIG. **41**) first by including an anvil face **1002** connecting to an anvil proximal end **1004** that includes laterally projecting anvil pivot pins **25** that are proximal to a vertically projecting anvil tab **27**. The anvil pivot pins **25** translate within kidney shaped openings **1006** in the staple channel **22** to open and close anvil **24** relative to channel **22**. The tab **27** engages a bent tab **1007** extending inwardly in tab opening **45** on a distal end **1008** of the closure tube **1005**, the latter distally terminating in a distal edge **1008** that pushes against the anvil face **1002**. Thus, when the closure tube **1005** moves proximally from its open position, the bent tab **1007** of the closure tube **1005** draws the anvil tab **27** proximally, and the anvil pivot pins **25** follow the kidney shaped openings **1006** of the staple channel **22** causing the anvil **24** to simultaneously translate proximally and rotate upward to the open position. When the closure tube **1005** moves distally, the bent tab **1007** in the tab opening **45** releases from the anvil tab **27** and the distal edge **1008** pushes on the anvil face **1002**, closing the anvil **24**.

(92) With continued reference to FIG. **41**, the shaft **8** and end effector **12** also include components that respond to a firing motion of a firing rod **1010**. In particular, the firing rod **1010** rotatably engages a firing trough member **1012** having a longitudinal recess **1014**. Firing trough member **1012** moves longitudinally within frame **1016** in direct response to longitudinal motion of firing rod **1010**. A longitudinal slot **1018** in the closure tube **1005** operably couples with the right and left exterior side handle pieces **61**, **62** of the handle **6** (not shown in FIG. **41**). The length of the longitudinal slot **1018** in the closure tube **1005** is sufficiently long to allow relative longitudinal motion with the handle pieces **61**, **62** to accomplish firing and closure motions respectively with the coupling of the handle pieces **61**, **62** passing on through a longitudinal slot **1020** in the frame **1016** to slidingly engage the longitudinal recess **1014** in the frame trough member **1012**.

(93) The distal end of the frame trough member **1012** is attached to a proximal end of a firing bar **1022** that moves within the frame **1016**, specifically within a guide **1024** therein, to distally project the knife **32** into the end effector **12**. The end effector **12** includes a staple cartridge **34** that is actuated by the knife **32**. The staple cartridge **34** has a tray **1028** that holds a staple cartridge body **1030**, a wedge sled driver **33**, staple drivers **1034** and staples **1036**. It will be appreciated that the wedge sled driver **33** longitudinally moves within a firing recess (not shown) located between the cartridge tray **1028** and the cartridge body **1030**. The wedge sled driver **33** presents camming surfaces that contact and lift the staple drivers **1034** upward, driving the staples **1036**. The staple cartridge body **1030** further includes a proximally open, vertical slot **1031** for passage of the knife **32**. Specifically, a cutting surface **1027** is provided along a distal end of knife **32** to cut tissue after it is stapled.

(94) It should be appreciated that the shaft **8** is shown in FIG. **4** as a non-articulating shaft. Nonetheless, applications of the present invention may include instruments capable of articulation,

for example, as such shown above with reference to FIGS. 1-4 and described in the following U.S. patents and patent applications, the disclosure of each being hereby incorporated by reference in their entirety: (1) SURGICAL INSTRUMENT INCORPORATING AN ARTICULATION MECHANISM HAVING ROTATION ABOUT THE LONGITUDINAL AXIS, U.S. Pat. No. 7,111,769; (2) SURGICAL STAPLING INSTRUMENT INCORPORATING AN ARTICULATION JOINT FOR A FIRING BAR TRACK, U.S. Pat. No. 6,786,382; (3) A SURGICAL INSTRUMENT WITH A LATERAL-MOVING ARTICULATION CONTROL, U.S. Pat. No. 6,981,628; (4) SURGICAL STAPLING INSTRUMENT INCORPORATING A TAPERED FIRING BAR FOR INCREASED FLEXIBILITY AROUND THE ARTICULATION JOINT, U.S. Pat. No. 6,964,363; and (5) SURGICAL STAPLING INSTRUMENT HAVING ARTICULATION JOINT SUPPORT PLATES FOR SUPPORTING A FIRING BAR, U.S. Pat. No. 7,055,731.

(95) FIGS. 42-43 show an embodiment of the handle **6** that is configured for use in a mechanically actuated endocutter along with the embodiment of the shaft **8** and end effector **12** as shown above in FIG. 41. It will be appreciated that any suitable handle design may be used to mechanically close and fire the end effector **12**. In FIGS. 42-43, the handle **6** of the surgical stapling and severing instrument **10** includes a linked transmission firing mechanism **1060** that provides features such as increased strength, reduced handle size, minimized binding, etc.

(96) Closure of the end effector **12** (not shown in FIGS. 42-43) is caused by depressing the closure trigger **18** toward the pistol grip **26** of handle **6**. The closure trigger **18** pivots about a closure pivot pin **252** that is coupled to right and left exterior lower side pieces **59**, **60** the handle **6**, causing an upper portion **1094** of the closure trigger **18** to move forward. The closure tube **1005** receives this closure movement via the closure yoke **250** that is pinned to a closure link **1042** and to the upper portion **1094** of the closure trigger **18** respectively by a closure yoke pin **1044** and a closure link pin **1046**.

(97) In the fully open position of FIG. 42, the upper portion **1094** of the closure trigger **18** contacts and holds a locking arm **1048** of the pivoting closure release button **30** in the position shown. When the closure trigger **18** reaches its fully depressed position, the closure trigger **18** releases the locking arm **1048** and an abutting surface **1050** rotates into engagement with a distal rightward notch **1052** of the pivoting locking arm **1048**, holding the closure trigger **18** in this clamped or closed position. A proximal end of the locking arm **1048** pivots about a lateral pivotal connection **1054** with the pieces **59**, **60** to expose the closure release button **30**. An intermediate, distal side **1056** of the closure release button **30** is urged proximally by a compression spring **1058**, which is compressed between a housing structure **1040** and closure release button **30**. The result is that the closure release button **30** urges the locking arm **1048** counterclockwise (when viewed from the left) into locking contact with the abutting surface **1050** of closure trigger **18**, which prevents unclamping of closure trigger **18** when the linked transmission firing system **1040** is in an un-retracted condition.

(98) With the closure trigger **18** retracted and fully depressed, the firing trigger **20** is unlocked and may be depressed toward the pistol grip **26**, multiple times in this embodiment, to effect firing of the end effector **12**. As depicted, the linked transmission firing mechanism **1060** is initially retracted, urged to remain in this position by a combination tension/compression spring **1062** that is constrained within the pistol grip **26** of the handle **6**, with its nonmoving end **1063** connected to the pieces **59**, **60** and a moving end **1064** connected to a downwardly flexed and proximal, retracted end **1067** of a steel band **1066**.

(99) A distally-disposed end **1068** of the steel band **1066** is attached to a link coupling **1070** for structural loading, which in turn is attached to a front link **1072a** of a plurality of links **1072a-1072d** that form a linked rack **1074**. Linked rack **1074** is flexible yet has distal links that form a straight rigid rack assembly that may transfer a significant firing force through the firing rod **1010** in the shaft **6**, yet readily retract into the pistol grip **26** to minimize the longitudinal length of the handle **6**. It should be appreciated that the combination tension/compression spring **1062** increases the amount of firing travel available while essentially reducing the minimum length by half over a

single spring.

(100) The firing trigger **20** pivots about a firing trigger pin **96** that is connected to the handle pieces **59, 60**. An upper portion **228** of the firing trigger **20** moves distally about the firing trigger pin **96** as the firing trigger **20** is depressed towards pistol grip **26**, stretching a proximally placed firing trigger tension spring **222** proximally connected between the upper portion **228** of the firing trigger **20** and the pieces **59, 60**. The upper portion **228** of the firing trigger **20** engages the linked rack **1074** during each firing trigger depression by a traction biasing mechanism **1078** that also disengages when the firing trigger **20** is released. Firing trigger tension spring **222** urges the firing trigger **20** distally when released and disengages the traction biasing mechanism **1078**.

(101) As the linked transmission firing mechanism **1040** actuates, an idler gear **1080** is rotated clockwise (as viewed from the left side) by engagement with a toothed upper surface **1082** of the linked rack **1074**. This rotation is coupled to an indicator gear **1084**, which thus rotates counterclockwise in response to the idler gear **1080**. Both the idler gear **1080** and indicator gear **1084** are rotatably connected to the pieces **59, 60** of the handle **6**. The gear relationship between the linked rack **1074**, idler gear **1080** and indicator gear **1084** may be advantageously selected so that the toothed upper surface **1082** has tooth dimensions that are suitably strong and that the indicator gear **1084** makes no more than one revolution during the full firing travel of the linked transmission firing mechanism **1060**.

(102) As described in greater detail below, the indicator gear **1084** performs at least four functions. First, when the linked rack **1074** is fully retracted and both triggers **18, 20** are open as shown in FIG. **42**, an opening **1086** in a circular ridge **1088** on the left side of the indicator gear **1084** is presented to an upper surface **1090** of the locking arm **1048**. Locking arm **1048** is biased into the opening **1086** by contact with the closure trigger **18**, which in turn is urged to the open position by a closure tension spring **1092**. Closure trigger tension spring **1092** is connected proximally to the upper portion **1094** of the closure trigger **18** and the handle pieces **59, 60**, and thus has energy stored during closing of the closure trigger **18** that urges the closure trigger **18** distally to its unclosed position.

(103) A second function of the indicator gear **1084** is that it is connected to the indicating retraction knob **1096** externally disposed on the handle **6**. Thus, the indicator gear **1084** communicates the relative position of the firing mechanism **1060** to the indicating retraction knob **1096** so that the surgeon has a visual indication of how many strokes of the firing trigger **20** are required to complete firing.

(104) A third function of the indicator gear **1084** is to longitudinally and angularly move an anti-backup release lever **1098** of an anti-backup mechanism (one-way clutch mechanism) **1097** as the surgical stapling and severing instrument **10** is operated. During the firing strokes, proximal movement of anti-backup release lever **1098** by indicator gear **1084** activates the anti-backup mechanism **1097** that allows distal movement of firing bar **1010** and prevents proximal motion of firing bar **1010**. This movement also extends the anti-backup release button **1100** from the proximal end of the handle pieces **59, 60** for the operator to actuate should the need arise for the linked transmission firing mechanism **1060** to be retracted during the firing strokes. After completion of the firing strokes, the indicator gear **1084** reverses direction of rotation as the firing mechanism **1060** retracts. The reversed rotation deactivates the anti-backup mechanism **1097**, withdraws the anti-backup release button **1100** into the handle **6**, and rotates the anti-backup release lever **1098** laterally to the right to allow continued reverse rotation of the indicator gear **1084**.

(105) A fourth function of the indicator gear **1084** is to receive a manual rotation from the indicating retraction knob **1096** (clockwise in the depiction of FIG. **42**) to retract the firing mechanism **1060** with anti-backup mechanism **1097** unlocked, thereby overcoming any binding in the firing mechanism **1060** that is not readily overcome by the combination tension/compression spring **1062**. This manual retraction assistance may be employed after a partial firing of the firing mechanism **1060** that would otherwise be prevented by the anti-backup mechanism **1097** that

withdraws the anti-backup release button **1100** so that the latter may not laterally move the anti-backup release lever **1098**.

(106) Continuing with FIGS. **42-43**, anti-backup mechanism **1097** consists of the operator accessible anti-backup release lever **1098** operably coupled at the proximal end to the anti-backup release button **1100** and at the distal end to an anti-backup yoke **1102**. In particular, a distal end **1099** of the anti-backup release lever **1098** is engaged to the anti-backup yoke **1102** by an anti-backup yoke pin **1104**. The anti-backup yoke **1102** moves longitudinally to impart a rotation to an anti-backup cam slot tube **1106** that is longitudinally constrained by the handle pieces **59, 90** and that encompasses the firing rod **1010** distally to the connection of the firing rod **1010** to the link coupling **1070** of the linked rack **1074**. The anti-backup yoke **1102** communicates the longitudinal movement from the anti-backup release lever **1098** via a cam slot tube pin **1108** to the anti-backup cam slot tube **1106**. That is, longitudinal movement of cam slot tube pin **1108** in an angled slot in the anti-backup cam slot tube **1106** rotates the anti-backup cam slot tube **1106**.

(107) Trapped between a proximal end of the frame **1016** and the anti-backup cam slot tube **1106** respectively are an anti-backup compression spring **1110**, an anti-backup plate **1112**, and an anti-backup cam tube **1114**. As depicted, proximal movement of the firing rod **1010** causes the anti-backup plate **1112** to pivot top to the rear, presenting an increased frictional contact to the firing rod **1010** that resists further proximal movement of the firing rod **1010**.

(108) This anti-backup plate **1112** pivots in a manner similar to that of a screen door lock that holds open a screen door when the anti-backup cam slot tube **1106** is closely spaced to the anti-backup cam tube **1114**. Specifically, the anti-backup compression spring **1110** is able to act upon a top surface of the plate **1112** to tip the anti-backup plate **1112** to its locked position. Rotation of the anti-backup cam slot tube **1106** causes a distal camming movement of the anti-backup cam tube **1114** thereby forcing the top of the anti-backup plate **1112** distally, overcoming the force from the anti-backup compression spring **1110**, thus positioning the anti-backup plate **1112** in an untipped (perpendicular), unlocked position that allows proximal retraction of the firing rod **1010**.

(109) With particular reference to FIG. **43**, the traction biasing mechanism **1078** is depicted as being composed of a pawl **1116** that has a distally projecting narrow tip **1118** and a rightwardly projecting lateral pin **1120** at its proximal end that is rotatably inserted through a hole **1076** in the upper portion **230** of the firing trigger **20**. On the right side of the firing trigger **20** the lateral pin **1120** receives a biasing member, depicted as biasing wheel **1122**. As the firing trigger **20** translates fore and aft, the biasing wheel **1122** traverses an arc proximate to the right half piece **59** of the handle **6**, overrunning at its distal portion of travel a biasing ramp **1124** integrally formed in the right half piece **59**. The biasing wheel **1122** may advantageously be formed from a resilient, frictional material that induces a counterclockwise rotation (when viewed from the left) into the lateral pin **1120** of the pawl **1116**, thus traction biasing the distally projecting narrow tip **1118** downward into a ramped central track **1075** of the nearest link **1072a-d** to engage the linked rack **1074**.

(110) As the firing trigger **20** is released, the biasing wheel **1122** thus tractionally biases the pawl **1116** in the opposite direction, raising the narrow tip **1118** from the ramped central track **1075** of the linked rack **1074**. To ensure disengagement of the tip **1118** under high load conditions and at nearly full distal travel of the pawl **1116**, the right side of the pawl **1116** ramps up onto a proximally and upwardly facing beveled surface **1126** on the right side of the closure yoke **250** to disengage the narrow tip **1118** from the ramped central track **1075**. If the firing trigger **20** is released at any point other than full travel, the biasing wheel **1122** is used to lift the narrow tip **1118** from the ramped central track **1075**. Whereas a biasing wheel **1122** is depicted, it should be appreciated that the shape of the biasing member or wheel **1122** is illustrative and may be varied to accommodate a variety of shapes that use friction or traction to engage or disengage the firing of the end effector **12**.

(111) Various embodiments of the surgical instrument **10** have the capability to record instrument

conditions at one or more times during use. FIG. 44 shows a block diagram of a system **2000** for recording conditions of the instrument **10**. It will be appreciated that the system **2000** may be implemented in embodiments of the instrument **10** having motorized or motor-assisted firing, for example, as described above with reference to FIGS. 1-40, as well as embodiments of the instrument **10** having mechanically actuated firing, for example, as described above with reference to FIGS. 41-43.

(112) The system **2000** may include various sensors **2002**, **2004**, **2006**, **2008**, **2010**, **2012** for sensing instrument conditions. The sensors may be positioned, for example, on or within the instrument **10**. In various embodiments, the sensors may be dedicated sensors that provide output only for the system **2000**, or may be dual-use sensors that perform other functions within the instrument **10**. For example, sensors **110**, **130**, **142** described above may be configured to also provide output to the system **2000**.

(113) Directly or indirectly, each sensor provides a signal to the memory device **2001**, which records the signals as described in more detail below. The memory device **2001** may be any kind of device capable of storing or recording sensor signals. For example, the memory device **2001** may include a microprocessor, an Electrically Erasable Programmable Read Only Memory (EEPROM), or any other suitable storage device. The memory device **2001** may record the signals provided by the sensors in any suitable way. For example, in one embodiment, the memory device **2001** may record the signal from a particular sensor when that signal changes states. In another embodiment, the memory device **2001** may record a state of the system **2000**, e.g., the signals from all of the sensors included in the system **2000**, when the signal from any sensor changes states. This may provide a snap-shot of the state of the instrument **10**. In various embodiments, the memory device **2001** and/or sensors may be implemented to include 1-WIRE bus products available from DALLAS SEMICONDUCTOR such as, for example, a 1-WIRE EEPROM.

(114) In various embodiments, the memory device **2001** is externally accessible, allowing an outside device, such as a computer, to access the instrument conditions recorded by the memory device **2001**. For example, the memory device **2001** may include a data port **2020**. The data port **2020** may provide the stored instrument conditions according to any wired or wireless communication protocol in, for example, serial or parallel format. The memory device **2001** may also include a removable medium **2021** in addition to or instead of the output port **2020**. The removable medium **2021** may be any kind of suitable data storage device that can be removed from the instrument **10**. For example, the removable medium **2021** may include any suitable kind of flash memory, such as a Personal Computer Memory Card International Association (PCMCIA) card, a COMPACTFLASH card, a MULTIMEDIA card, a FLASHMEDIA card, etc. The removable medium **2021** may also include any suitable kind of disk-based storage including, for example, a portable hard drive, a compact disk (CD), a digital video disk (DVD), etc.

(115) The closure trigger sensor **2002** senses a condition of the closure trigger **18**. FIGS. 45 and 46 show an exemplary embodiment of the closure trigger sensor **2002**. In FIGS. 45 and 46, the closure trigger sensor **2002** is positioned between the closure trigger **18** and closure pivot pin **252**. It will be appreciated that pulling the closure trigger **18** toward the pistol grip **26** causes the closure trigger **18** to exert a force on the closure pivot pin **252**. The sensor **2002** may be sensitive to this force, and generate a signal in response thereto, for example, as described above with respect to sensor **110** and FIGS. 10A and 10B. In various embodiments, the closure trigger sensor **2002** may be a digital sensor that indicates only whether the closure trigger **18** is actuated or not actuated. In other various embodiments, the closure trigger sensor **2002** may be an analog sensor that indicates the force exerted on the closure trigger **18** and/or the position of the closure trigger **18**. If the closure trigger sensor **2002** is an analog sensor, an analog-to-digital converter may be logically positioned between the sensor **2002** and the memory device **2001**. Also, it will be appreciated that the closure trigger sensor **2002** may take any suitable form and be placed at any suitable location that allows sensing of the condition of the closure trigger.

(116) The anvil closure sensor **2004** may sense whether the anvil **24** is closed. FIG. 47 shows an exemplary anvil closure sensor **2004**. The sensor **2004** is positioned next to, or within the kidney shaped openings **1006** of the staple channel **22** as shown. As the anvil **24** is closed, anvil pivot pins **25** slides through the kidney shaped openings **1006** and into contact with the sensor **2004**, causing the sensor **2004** to generate a signal indicating that the anvil **24** is closed. The sensor **2004** may be any suitable kind of digital or analog sensor including a proximity sensor, etc. It will be appreciated that when the anvil closure sensor **2004** is an analog sensor, an analog-to-digital converter may be included logically between the sensor **2004** and the memory device **2001**.

(117) Anvil closure load sensor **2006** is shown placed on an inside bottom surface of the staple channel **22**. In use, the sensor **2006** may be in contact with a bottom side of the staple cartridge **34** (not shown in FIG. 46). As the anvil **24** is closed, it exerts a force on the staple cartridge **34** which is transferred to the sensor **2006**. In response, the sensor **2006** generates a signal. The signal may be an analog signal proportional to the force exerted on the sensor **2006** by the staple cartridge **34** and due to the closing of the anvil **24**. Referring the FIG. 44, the analog signal may be provided to an analog-to-digital converter **2014**, which converts the analog signal to a digital signal before providing it to the memory device **2001**. It will be appreciated that embodiments where the sensor **2006** is a digital or binary sensor may not include analog-to-digital converter **2014**.

(118) The firing trigger sensor **110** senses the position and/or state of the firing trigger **20**. In motorized or motor-assisted embodiments of the instrument, the firing trigger sensor may double as the run motor sensor **110** described above. In addition, the firing trigger sensor **110** may take any of the forms described above, and may be analog or digital. FIGS. 45 and 46 show an additional embodiment of the firing trigger sensor **110**. In FIGS. 45 and 46, the firing trigger sensor is mounted between firing trigger **20** and firing trigger pivot pin **96**. When firing trigger **20** is pulled, it will exert a force on firing trigger pivot pin **96** that is sensed by the sensor **110**. Referring to FIG. 44, in embodiments where the output of the firing trigger sensor **110** is analog, analog-to-digital converter **2016** is included logically between the firing trigger sensor **110** and the memory device **2001**.

(119) The knife position sensor **2008** senses the position of the knife **32** or cutting surface **1027** within the staple channel **22**. FIGS. 47 and 48 show embodiments of a knife position sensor **2008** that are suitable for use with the mechanically actuated shaft **8** and end effector **12** shown in FIG. 41. The sensor **2008** includes a magnet **2009** coupled to the firing bar **1022** of the instrument **10**. A coil **2011** is positioned around the firing bar **1022**, and may be installed; for example, along the longitudinal recess **1014** of the firing trough member **1012** (see FIG. 41). As the knife **32** and cutting surface **1027** are reciprocated through the staple channel **22**, the firing bar **1022** and magnet **2009** may move back and forth through the coil **2011**. This motion relative to the coil induces a voltage in the coil proportional to the position of the firing rod within the coil and the cutting edge **1027** within the staple channel **22**. This voltage may be provided to the memory device **2001**, for example, via analog-to-digital converter **2018**.

(120) In various embodiments, the knife position sensor **2008** may instead be implemented as a series of digital sensors (not shown) placed at various positions on or within the shaft **8**. The digital sensors may sense a feature of the firing bar **1022** such as, for example, magnet **2009**, as the feature reciprocates through the shaft **8**. The position of the firing bar **1022** within the shaft **8**, and by extension, the position of the knife **32** within the staple channel **22**, may be approximated as the position of the last digital sensor tripped.

(121) It will be appreciated that the knife position may also be sensed in embodiments of the instrument **10** having a rotary driven end effector **12** and shaft **8**, for example, as described above, with reference to FIGS. 3-6. An encoder, such as encoder **268**, may be configured to generate a signal proportional to the rotation of the helical screw shaft **36**, or any other drive shaft or gear. Because the rotation of the shaft **36** and other drive shafts and gears is proportional to the movement of the knife **32** through the channel **22**, the signal generated by the encoder **268** is also

proportional to the movement of the knife **32**. Thus, the output of the encoder **268** may be provided to the memory device **2001**.

(122) It will be appreciated that the position of an actuator that drives a surgical function of the end effector **12** may also be sensed on powered embodiments of the instrument **10**, as described above, for example, with reference to FIGS. **3-10B**. Actuators, or moving elements, may drive surgical functions of the end effector **12** such as, for example, cutting, stapling, closure, and articulation. When a position sensor senses that an actuator, such as, for example, firing bar **1022** or helical screw shaft **36**, is at a position indicative that the surgical function is at an end-of-stroke or at a limit position, the position sensor signals a microprocessor in the force delivery system that the limit has been reached. The microprocessor, such as one that may be included in memory device **2001**, can cause a change in voltage driving the motor. The change in voltage may increase, decrease, or reverse the motor's speed of rotation in a manner to cause a change in force, and thus a force feedback, to a user holding the handle. The user then senses that the surgical function is at the limit position. Alternately, the signal from the position sensor, the voltage change, and the force feedback may occur when the surgical function is approaching the limit position without having yet achieved the limit position.

(123) The force delivered to the user may be a vibratory force. FIGS. **54** and **55** demonstrate a changing voltage, and a changing force, that is delivered to the user in the case of a vibratory feedback showing end-of-stroke for an actuator causing articulation. If an end effector such as end effector **12** is articulated to the right to its articulation limit, a position sensor can sense actuator movement corresponding to the limit position and cause a microprocessor to send a voltage to the motor driving articulation. The signal can comprise a rapidly varying square wave or sine wave to drive the motor in a vibratory manner sensible by the user holding the handle. Likewise, an end-effector articulated to the left to its articulation limit can also be signaled.

(124) FIG. **56** shows a flow diagram illustrating processor logic that can be repeated multiple times per second to determine if an actuator causing articulation can be advanced further or if vibratory feedback is warranted. A processor following this logic would begin at **3005** at the initiation of articulation. If a user depresses a switch such as switch **110** to initiate articulation to the right, the processor logic, at decision point **3025**, uses a signal from a position sensor to check to see if end effector **12** is at the right limit. If the end effector **12** is at the right limit, the processor logic proceeds to state **3020**, where the processor sends the rapidly varying square or sign wave. If the end effector **12** is not at the right limit, the processor logic proceeds to **3030**, allowing further articulation until the user ends articulation or until the right limit is reached. Processor logic proceeds similarly to articulate the end effector **12** to the extreme left limit. Similar logic may also be used with motors and actuators that drive the surgical functions of cutting, stapling, closure, and opening.

(125) The cartridge present sensor **2010** may sense the presence of the staple cartridge **34** within the staple channel **22**. In motorized or motor-assisted instruments, the cartridge present sensor **2010** may double as the cartridge lock-out sensor **136** described above with reference to FIG. **11**. FIGS. **50** and **51** show an embodiment of the cartridge present sensor **2010**. In the embodiment shown, the cartridge present sensor **2010** includes two contacts, **2011** and **2013**. When no cartridge **34** is present, the contacts **2011**, **2013** form an open circuit. When a cartridge **34** is present, the cartridge tray **1028** of the staple cartridge **34** contacts the contacts **2011**, **2013**, a closed circuit is formed. When the circuit is open, the sensor **2010** may output a logic zero. When the circuit is closed, the sensor **2010** may output a logic one. The output of the sensor **2010** is provided to memory device **2001**, as shown in FIG. **44**.

(126) The cartridge condition sensor **2012** may indicate whether a cartridge **34** installed within the staple channel **22** has been fired or spent. As the knife **32** is translated through the end effector **12**, it pushes the sled **33**, which fires the staple cartridge. Then the knife **32** is translated back to its original position, leaving the sled **33** at the distal end of the cartridge. Without the sled **33** to guide

it, the knife **32** may fall into lock-out pocket **2022**. Sensor **2012** may sense whether the knife **32** is present in the lock-out pocket **2022**, which indirectly indicates whether the cartridge **34** has been spent. It will be appreciated that in various embodiments, sensor **2012** may directly sense the present of the sled at the proximate end of the cartridge **34**, thus eliminating the need for the knife **32** to fall into the lock-out pocket **2022**.

(127) FIGS. **52A** and **52B** depict a process flow **2200** for operating embodiments of the surgical instrument **10** configured as an endocutter and having the capability to record instrument conditions according to various embodiments. At box **2202**, the anvil **24** of the instrument **10** may be closed. This causes the closure trigger sensor **2002** and/or the anvil closure sensor **2006** to change state. In response, the memory device **2001** may record the state of all of the sensors in the system **2000** at box **2203**. At box **2204**, the instrument **10** may be inserted into a patient. When the instrument is inserted, the anvil **24** may be opened and closed at box **2206**, for example, to manipulate tissue at the surgical site. Each opening and closing of the anvil **24** causes the closure trigger sensor **2002** and/or the anvil closure sensor **2004** to change state. In response, the memory device **2001** records the state of the system **2000** at box **2205**.

(128) At box **2208**, tissue is clamped for cutting and stapling. If the anvil **24** is not closed at decision block **2210**, continued clamping is required. If the anvil **24** is closed, then the sensors **2002**, **2004** and/or **2006** may change state, prompting the memory device **2001** to record the state of the system at box **2213**. This recording may include a closure pressure received from sensor **2006**. At box **2212**, cutting and stapling may occur. Firing trigger sensor **110** may change state as the firing trigger **20** is pulled toward the pistol grip **26**. Also, as the knife **32** moves through the staple channel **22**, knife position sensor **2008** will change state. In response, the memory device **2001** may record the state of the system **2000** at box **2013**.

(129) When the cutting and stapling operations are complete, the knife **32** may return to a pre-firing position. Because the cartridge **34** has now been fired, the knife **32** may fall into lock-out pocket **2022**, changing the state of cartridge condition sensor **2012** and triggering the memory device **2001** to record the state of the system **2000** at box **2015**. The anvil **24** may then be opened to clear the tissue. This may cause one or more of the closure trigger sensor **2002**, anvil closure sensor **2004** and anvil closure load sensor **2006** to change state, resulting in a recordation of the state of the system **2000** at box **2017**. After the tissue is cleared, the anvil **24** may be again closed at box **2220**. This causes another state change for at least sensors **2002** and **2004**, which in turn causes the memory device **2001** to record the state of the system at box **2019**. Then the instrument **10** may be removed from the patient at box **2222**.

(130) If the instrument **10** is to be used again during the same procedure, the anvil may be opened at box **2224**, triggering another recordation of the system state at box **2223**. The spent cartridge **34** may be removed from the end effector **12** at box **2226**. This causes cartridge present sensor **2010** to change state and cause a recordation of the system state at box **2225**. Another cartridge **34** may be inserted at box **2228**. This causes a state change in the cartridge present sensor **2010** and a recordation of the system state at box **2227**. If the other cartridge **34** is a new cartridge, indicated at decision block **2230**, its insertion may also cause a state change to cartridge condition sensor **2012**. In that case, the system state may be recorded at box **2231**.

(131) FIG. **53** shows an exemplary memory map **2300** from the memory device **2001** according to various embodiments. The memory map **2300** includes a series of columns **2302**, **2304**, **2306**, **2308**, **2310**, **2312**, **2314**, **2316** and rows (not labeled). Column **2302** shows an event number for each of the rows. The other columns represent the output of one sensor of the system **2000**. All of the sensor readings recorded at a given time may be recorded in the same row under the same event number. Hence, each row represents an instance where one or more of the signals from the sensors of the system **2000** are recorded.

(132) Column **2304** lists the closure load recorded at each event. This may reflect the output of anvil closure load sensor **2006**. Column **2306** lists the firing stroke position. This may be derived

from the knife position sensor **2008**. For example, the total travel of the knife **32** may be divided into partitions. The number listed in column **2306** may represent the partition where the knife **32** is currently present. The firing load is listed in column **2308**. This may be derived from the firing trigger sensor **110**. The knife position is listed at column **2310**. The knife position may be derived from the knife position sensor **2008** similar to the firing stroke. Whether the anvil **24** is open or closed may be listed at column **2312**. This value may be derived from the output of the anvil closure sensor **2004** and/or the anvil closure load sensor **2006**. Whether the sled **33** is present, or whether the cartridge **34** is spent, may be indicated at column **2314**. This value may be derived from the cartridge condition sensor **2012**. Finally, whether the cartridge **34** is present may be indicated a column **2316**. This value may be derived from cartridge present sensor **2010**. It will be appreciated that various other values may be stored at memory device **2001** including, for example, the end and beginning of firing strokes, for example, as measured by sensors **130**, **142**.

(133) While the present invention has been illustrated by description of several embodiments and while the illustrative embodiments have been described in considerable detail, it is not the intention of the applicant to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications may readily appear to those skilled in the art.

(134) For example, although the embodiments described above have advantages for an endoscopically employed surgical severing and stapling instrument **100**, a similar embodiments may be used in other clinical procedures. It is generally accepted that endoscopic procedures are more common than laparoscopic procedures. Accordingly, the present invention has been discussed in terms of endoscopic procedures and apparatus. However, use herein of terms such as “endoscopic”, should not be construed to limit the present invention to a surgical instrument for use only in conjunction with an endoscopic tube (i.e., trocar). On the contrary, it is believed that the present invention may find use in any procedure where access is limited to a small incision, including but not limited to laparoscopic procedures, as well as open procedures.

(135) Any patent, publication, or information, in whole or in part, that is said to be incorporated by reference herein is incorporated herein only to the extent that the incorporated material does not conflict with existing definitions, statements, or other disclosure material set forth in this document. As such the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference.

Claims

1. A surgical assembly, comprising: a shaft; an end effector; a driver actuatable by a motor, wherein the driver is movable through a drive motion, and wherein the end effector comprises a first jaw and a second jaw movable relative to the first jaw; and a control circuit communicably coupled to the motor, the control circuit to: receive a user input corresponding to an initiation of the drive motion; initiate the drive motion based on the user input; continuously monitor the position of the driver throughout the drive motion; and deliver a feedback to a user over a period of time based on the position of the driver, wherein the feedback comprises an indication that the driver is approaching the conclusion of the drive motion.
2. The surgical assembly of claim 1, wherein the feedback comprises a tactile indication that the driver is approaching the conclusion of the drive motion.
3. The surgical assembly of claim 2, further comprising a motor to generate the tactile indication that the driver is approaching the conclusion of the drive motion.
4. The surgical assembly of claim 1, wherein the control circuit is further to transmit a modulating signal profile based on the position of the driver.
5. The surgical assembly of claim 4, wherein the modulating signal profile corresponds to a haptic feedback to the user.
6. The surgical assembly of claim 1, wherein the feedback comprises an indication that the driver is

approaching a limit position of the drive motion.

7. The surgical assembly of claim 6, further comprising an articulation joint, wherein the driver comprises an articulation cable, and wherein the drive motion comprises an articulation motion.

8. The surgical assembly of claim 1, wherein the driver comprises an I-beam.

9. The surgical assembly of claim 8, wherein the drive motion comprises a closure stroke.

10. The surgical assembly of claim 8, wherein the drive motion comprises a firing stroke.

11. A surgical assembly, comprising: a shaft; an end effector; a driver actuatable by a motor, wherein the driver is movable through a drive motion, and wherein the end effector comprises a first jaw and a second jaw movable relative to the first jaw; and a control circuit communicably coupled to the motor, the control circuit to: receive a user input corresponding to an initiation of the drive motion; initiate the drive motion based on the user input; continuously monitor the status of the driver throughout the drive motion; and deliver a feedback to a user over a period of time based on the status of the driver, wherein the feedback comprises an indication that the driver is approaching the conclusion of the drive motion.

12. The surgical assembly of claim 11, wherein the feedback comprises a tactile indication that the driver is approaching the conclusion of the drive motion.

13. The surgical assembly of claim 12, further comprising a motor to generate the tactile indication that the driver is approaching the conclusion of the drive motion.

14. The surgical assembly of claim 11, wherein the control circuit is further to transmit a modulating signal profile based on the status of the driver.

15. The surgical assembly of claim 14, wherein the modulating signal profile corresponds to a haptic feedback to the user.

16. The surgical assembly of claim 11, wherein the feedback comprises an indication that the driver is approaching a limit position of the drive motion.

17. The surgical assembly of claim 16, wherein the driver comprises an I-beam.

18. The surgical assembly of claim 17, wherein the drive motion comprises a closure stroke.

19. The surgical assembly of claim 17, wherein the drive motion comprises a firing stroke.

20. The surgical assembly of claim 11, further comprising an articulation joint, wherein the driver comprises an articulation cable, wherein the drive motion comprises an articulation motion, and wherein the status of the driver comprises the position of the articulation cable throughout the articulation motion.

21. A surgical assembly, comprising: a shaft; an end effector, wherein the end effector comprises a first jaw and a second jaw movable relative to the first jaw; a motorized driver movable through a motorized drive motion; and a control circuit to: initiate the motorized drive motion; continuously track a position of the motorized driver throughout the motorized drive motion; and deliver a feedback force to a user during the motorized drive motion based on the position of the motorized driver.

22. The surgical assembly of claim 21, wherein the motorized driver comprises an I-beam.

23. The surgical assembly of claim 22, wherein the motorized drive motion comprises a closure stroke.

24. The surgical assembly of claim 23, further comprising a sensor to detect the position of the motorized driver.

25. The surgical assembly of claim 22, wherein the motorized drive motion comprises a firing stroke.

26. The surgical assembly of claim 25, further comprising a sensor to detect the position of the motorized driver.

27. The surgical assembly of claim 26, wherein the feedback force comprises a vibration.

28. The surgical assembly of claim 27, further comprising a motor to generate the vibration.

29. The surgical assembly of claim 21, wherein the feedback force comprises an indication that the driver is approaching a limit position of the motorized drive motion.

30. The surgical assembly of claim 21, further comprising an articulation joint, wherein the driver comprises an articulation cable, and wherein the motorized drive motion comprises an articulation motion.
